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Experiential Learning Utility Report

Birthday Reminder

Course: Python Programming Lab (N-PCCCD304P)
Branch / Semester: CSE (DS) / III

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GitHub Repository	(paste your repository link here)

1. Problem Statement

Develop a Python application that stores birthday information of friends, family, or employees and reminds users of upcoming birthdays. The application should support adding, updating, searching, deleting, and notifying users about birthdays.

2. Objectives

• To store birthday details (name, date, category, phone) permanently in a file.
• To add, update, search and delete birthday records.
• To find the next birthday of every person and show the days remaining.
• To notify the user about birthdays that are today and in the next 7 days.
• To show simple statistics such as average age and busiest birthday month.
• To apply all Python Programming Lab modules in one practical utility.
• To manage the project with Git and GitHub using proper commits and README.

3. Modules and Python Concepts Used

Module	Topic	Where used in the utility
1	Fundamentals / Control structures	Menu with while loop, if-elif-else, for loops, input/output
2	Data structures	String (name, date), list (records), tuple (categories), set (unique months), dictionary (month count)
3	Functions	parse_date(), next_birthday(), days_until(), age_turning(), ask_date(), write_log()
4	NumPy, Pandas	Pandas reads/writes CSV and counts categories; NumPy finds mean, min, max age
5	File handling and OOP	CSV data file, text log file; classes Person and BirthdayManager

4. Algorithm

1. Start the program and load all records from birthdays.csv (if the file exists).
2. Show today's and the next 7 days' birthday notifications.
3. Display the menu and read the user's choice.
4. **Add:** read name, date (DD-MM-YYYY), category and phone; validate; add to list; save to file.
5. **Update:** find the person by name; change only the fields the user enters; save.
6. **Search:** compare the keyword with the name (part of name) or category; show matches.
7. **Delete:** find the person, ask confirmation, remove from list; save.
8. **Next birthday:** put the birth day and month in the current year; if that date has passed, use next year.
9. **Days left** = next birthday - today. Sort by days left for the upcoming list. Days left = 0 means birthday today.
10. Repeat the menu until the user chooses Exit.

5. Flow of the Program

```
START
  |
Load records from CSV
  |
Show notifications
  |
+--> Show menu --> Read choice
|      |
|      +-- 1 Add      --> validate --> save
|      +-- 2 Update   --> find --> change --> save
|      +-- 3 Search   --> show matches
|      +-- 4 Delete   --> confirm --> save
|      +-- 5-8 View / Upcoming / Notify / Statistics
|      +-- 9 Exit     --> STOP
+------(repeat)
```

6. Classes and Functions

Name	Purpose
Person (class)	Stores name, birth date, category, phone of one person; to_dict() and __str__()
BirthdayManager (class)	Holds the list of people; add_person, update_person, search, delete_person, upcoming, notify, statistics, load, save
parse_date()	Converts text to a date and rejects invalid or future dates
next_birthday()	Returns the next birthday date (handles 29 Feb)
days_until()	Days left for the next birthday
age_turning()	Age the person will turn on the next birthday
is_valid_phone()	Phone is optional; must be 10 digits if given
write_log()	Appends a time-stamped line to reminder_log.txt

7. Output Screenshots

Note: these outputs are sample runs of the program. Replace them with your own terminal screenshots after running the program on your computer.

7.1 Startup notification

```
*** BIRTHDAY NOTIFICATIONS ***
TODAY: It's Rohan Deshmukh's birthday! (turns 22)
```

7.2 View all birthdays (nearest first)

```
All birthdays
Name               Birthday    Category  Phone
-------------------------------------------------
Rohan Deshmukh     21-09-2004  Friend    9823456781
Sneha Kulkarni     30-09-2005  Friend    -
Priya Borikar      08-10-1978  Family    9765432109
Amit Sharma        12-12-1990  Employee  9890123456
Asha Patil         15-03-2005  Friend    9876543210
Meera Joshi        25-06-2003  Family    9922334455
```

7.3 Search (keyword: asha)

```
Enter name or category to search: asha
Search results
Name               Birthday    Category  Phone
-----------------------------------------------
Asha Patil         15-03-2005  Friend    9876543210
```

7.4 Add a birthday

```
Name: Kiran Rao
Birthday (DD-MM-YYYY): 10-05-2004
Category:
  1. Friend
  2. Family
  3. Employee
Choose 1-3: 1
Phone (10 digits, optional): 9812345678
Kiran Rao added successfully.
```

7.5 Update a birthday

```
Name of the person to update: Kiran Rao
Leave blank to keep the old value.
New name:
New birthday (DD-MM-YYYY): 11-05-2004
Change category? (y/n): n
New phone (blank = keep):
Record updated.
```

7.6 Delete a birthday

```
Name to delete: Kiran Rao
Delete 'Kiran Rao'? (y/n): y
Kiran Rao deleted.
```

7.7 Statistics

```
--- Statistics ---
Total records   : 6
Average age     : 28.0
Youngest / Oldest: 20 / 47

People per category:
Category
Friend      3
Family      2
Employee    1

Busiest birthday month: September (2)
Months with birthdays: 5 of 12
```

7.8 Invalid date and invalid phone

```
Name: Asha Patil
Birthday (DD-MM-YYYY): 31-02-2005
Invalid date. Use DD-MM-YYYY (past date), e.g. 25-12-2005.
Birthday (DD-MM-YYYY): 15-03-2005
Category: 1
Phone (10 digits, optional): 123
Phone must be 10 digits (or leave empty).
```

7.9 Duplicate name

```
Name: Asha Patil
Birthday (DD-MM-YYYY): 15-03-2005
Category: 1
Phone (10 digits, optional):
This name already exists.
```

8. Test Cases

No.	Input	Expected result	Result
1	Add Asha Patil, 15-03-2005, Friend	Record added successfully	Pass
2	Add the same name again	This name already exists.	Pass
3	Date 31-02-2005	Invalid date, asks again	Pass
4	Phone 123	Phone must be 10 digits	Pass
5	Search asha	Shows Asha Patil's record	Pass
6	Search Family	Shows all Family records	Pass
7	Delete an unknown name	Person not found.	Pass
8	Birthday today	TODAY message shown at start	Pass
9	Menu choice 15	Invalid choice message	Pass
10	Close and reopen program	Old records still available	Pass

9. Version Control (Git / GitHub)

- Public repository with main branch as the stable version.
- Files: birthday_reminder.py, README.md, requirements.txt, birthdays.csv, Report.pdf.
- Feature branch used for development and merged into main using a pull request.
- Clear commit messages describing each change.

10. Conclusion

The Birthday Reminder utility successfully stores, updates, searches, deletes and reminds users about birthdays. It uses variables, control structures, data structures, functions, NumPy, Pandas, file handling and object-oriented programming, so it covers all five modules of the Python Programming Lab. In future it can be extended with email or SMS reminders and a graphical interface.